Pseudo-Rational Choice with Multiple Tastes and Beliefs

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Abstract

This paper aims to provide an axiomatic characterization of pseudo-rational multi-taste, multi-prior choice over menus of Anscombe–Aumann acts. Taking a choice correspondence as primitive, the framework allows for violations of the Weak Axiom of Revealed Preference. The main objective is to characterize choice correspondences admitting a representation of the form

$$c(A) = \bigcup_{(\pi,u)\in\mathcal{M}} \arg\max_{f\in A} \sum_{s\in S} \pi(s)\mathbb{E}_{f(s)}[u],$$

where $\mathcal{M}$ is a set of prior–utility pairs, with each prior representing beliefs over states and each utility function representing a possible evaluation of prizes.

The model follows the notion of justifiable choice: an act is chosen whenever it maximizes subjective expected utility under at least one admissible belief–taste pair. This allows for uncertainty regarding both the states of the world and the decision maker's tastes.

The analysis focuses initially on finite prize spaces and explores an extension to compact metric prize spaces. The proposed approach draws on maximal-menu arguments and Heller's (2012) minimax theorem method to investigate the connection between revealed choice and multi-utility representations. Finally, the paper seeks to explore how indecisiveness and indifference may be identified and distinguished from observed choice behavior.

Keywords: Menu effects, Multiple priors, Multiple utilities, Incomplete preferences, Revealed preference, Indecisiveness.

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1 Introduction

The literature studying indecisiveness in beliefs and tastes using a binary relation as the primitive is recent but includes several notable contributions (Riella, 2014; Hara and Riella, 2015; Galaabaatar and Karni, 2013). By contrast, comparatively little work addresses these questions using a choice correspondence.

Eliaz and Ok (2006) provide a pioneering contribution by introducing the WARNI axiom and characterizing when observed behavior can distinguish indecisiveness from indifference. More recently, Heller (2012) and Safonov (2023) study this distinction in environments involving risk and uncertainty, offering insights into indecisiveness in either tastes or beliefs. Their models implicitly treat the decision maker's behavior as pseudo-rational choice in the sense of Aizerman and Malishevski (1981).

To the best of our knowledge, however, no model characterizes conditions on a choice correspondence under which a decision maker behaves as if she were simultaneously indecisive about both her tastes and her beliefs, rather than about either dimension separately.

2 Model

We consider an Anscombe–Aumann setup with a finite state space S and a compact metric prize space X. An act is a function that maps each state of the world $s \in S$ to a lottery $f(s) \in \Delta(X)$, and we denote the set of all acts by $\mathcal{F}$.

Here, $\Delta(X)$ denotes the set of Borel probability measures on X, endowed with any metric that metrizes the topology of weak convergence, and $\mathcal{F} := \Delta(X)^S$ is endowed with the product metric. Thus, $f_n \to f$ in $\mathcal{F}$ if and only if $f_n(s)$ converges weakly to $f(s)$ for every $s \in S$. Since X is compact and metrizable and S is finite, both $\Delta(X)$ and $\mathcal{F}$ are compact metrizable spaces. Whenever continuous utility functions are considered, $C(Y)$ denotes the space of continuous real-valued functions on a compact metric space Y, endowed with the supremum norm. In particular, state-dependent utility functions belong to $C(X \times S)$, while state-independent utility functions belong to $C(X)$.

Let $\mathcal{A}$ be the collection of nonempty compact menus of acts, endowed with the Hausdorff metric induced by the metric on $\mathcal{F}$, so that $\mathcal{A} \subseteq 2^{\mathcal{F}} \setminus \{\emptyset\}$. Our primitive is a choice correspondence $c\colon \mathcal{A} \to 2^{\mathcal{F}} \setminus \{\emptyset\}$ such that, for every $A \in \mathcal{A}$, $\emptyset \neq c(A) \subseteq A$. The choice correspondence is an object that the analyst can usually observe and is therefore more primitive than the notion of preference (Ok and Eliaz, 2006). Its interpretation is as follows: when the decision maker is presented with a menu A, she may choose any act in $c(A)$.

We define mixtures of acts state by state. That is, for every $\alpha \in [0,1]$, every $g, h \in \mathcal{F}$, and every $s \in S$, $\big(\alpha g + (1-\alpha)h\big)(s) = \alpha g(s) + (1-\alpha)h(s)$. Hence, $\mathcal{F}$ can also be viewed as a mixture space. We abuse notation by using $x \in X$ to denote the degenerate constant act that yields outcome x with certainty in every state, and $p \in \Delta(X)$ to denote the constant act that yields lottery p in every state $s \in S$. For $\nu \in C(X \times S)$ and $f \in \mathcal{F}$, the associated state-dependent expected utility is

$$U_\nu(f) := \sum_{s \in S} \int_X \nu(x,s)\, df(s)(x).$$

For a prior $\pi \in \Delta(S)$ and a state-independent utility function $u \in C(X)$, define the subjective expected utility of $f \in \mathcal{F}$ by

$$V_{\pi,u}(f) := \sum_{s \in S} \pi(s) \int_X u(x)\, df(s)(x).$$

Here, $\Delta(S)$ is the simplex of probability distributions on S, endowed with its Euclidean topology. The space $\Delta(S) \times C(X)$ carries the product of this topology and the supremum-norm topology. Since u is continuous and X is compact, the integrals are well defined and $V_{\pi,u}$ is continuous and affine on $\mathcal{F}$. For a constant act $p \in \Delta(X)$, $V_{\pi,u}(p) = \int_X u(x)\, dp(x)$. When X is finite, both U_ν and $V_{\pi,u}$ reduce to the corresponding finite sums.

2.1 Axioms

We impose the following axioms on the choice correspondence c.

Axiom 1 (Pseudo-WARP). For all $A, B \in \mathcal{A}$, if $f \in c(B) \cap A$ and $c(A) \subseteq B$, then $f \in c(A)$.

Axiom 2 (Independence). For every $A \in \mathcal{A}$, $\lambda \in (0,1)$, and $h \in \mathcal{F}$,[1] $c\big(\lambda A + (1-\lambda)h\big) = \lambda c(A) + (1-\lambda)h$.

Axiom 3 (Continuity). The set $\big\{(A, f) \in \mathcal{A} \times \mathcal{F} : f \in c(A)\big\}$ is closed.

Axiom 4 (Indirect Ambiguity Aversion). For every $A \in \mathcal{A}$, $\lambda \in [0,1]$, and $f, g, h \in \mathcal{F}$, if $f \notin c\big(A \cup \{h\}\big)$ and $f \notin c\big(A \cup \{g\}\big)$, then $f \notin c\big(A \cup \{\lambda g + (1-\lambda)h\}\big)$.

Axiom 5 (No-Hedging). For every $A \in \mathcal{A}$, $\lambda \in [0,1]$, and $g, h \in \mathcal{F}$, if $\{g, h\} \subseteq A$, then $c(A) \subseteq c\big(A \cup \{\lambda g + (1-\lambda)h\}\big)$.

Axiom 6 (Constant Non-Triviality). There exist $\hat{p}, \hat{q} \in \Delta(X)$ such that $\hat{q} \notin c\big(\{\hat{p}, \hat{q}\}\big)$.

Axiom 7 (Strict One-Side Monotonicity). For every $A \in \mathcal{A}$ and every $f, g \in \mathcal{F}$, if $f \notin c\big(A \cup \{g(s)\}\big)$ for every $s \in S$, then $f \notin c\big(A \cup \{g\}\big)$.

Pseudo-WARP imposes consistency across menus while allowing the choice correspondence to be generated by multiple possible rationales. Independence requires choice to commute with a common mixture transformation, and Continuity rules out discontinuous changes in choice as menus and acts converge. Indirect Ambiguity Aversion imposes a form of consistency under hedging: if f is not chosen in the presence of h, and h is not chosen in the presence of g, then replacing h with any mixture of g and h cannot make f chosen.

No-Hedging requires that adding a mixture of two acts already available in a menu does not eliminate any act that was previously chosen. Constant Non-Triviality requires that there exist two constant acts $\hat{p}, \hat{q} \in \Delta(X)$ such that $\hat{q}$ is not chosen from $\{\hat{p}, \hat{q}\}$. Finally, Strict One-Side Monotonicity requires that if f is not chosen when each statewise lottery of g is added separately to a menu as a constant act, then f is not chosen when g itself is added to that menu.

[1] We abuse notation and write $\lambda A + (1-\lambda)h$ for the set $\{\lambda f + (1-\lambda)h : f \in A\}$.

3 Representation

3.1 Additively Separable Expected Multi-Utility Representation

This subsection axiomatizes state-dependent expected multi-utility choice for a compact metric prize space X. Related representations with a preorder as the primitive are obtained by Ok et al. (2012) under Independence and Continuity for finite prize spaces and extended by Riella (2015) to compact metric spaces.

Our choice-based analysis requires an auxiliary representation of a strict partial order by a compact, convex set of continuous state-dependent utilities. While Galaabaatar and Karni (2012) study related representations in finite-dimensional spaces, our setting allows a compact metric prize space. Proposition 1 establishes the required result by adapting the approach of Hara and Riella (2023).

Consider a strict relation $\succ \subseteq \mathcal{F} \times \mathcal{F}$ and the following axioms:

(S1) Independence. For all $f, g, h \in \mathcal{F}$ and $\alpha \in (0, 1]$, $f \succ g \iff \alpha f + (1-\alpha)h \succ \alpha g + (1-\alpha)h$.

(S2) Open Continuity. For every $f \in \mathcal{F}$, the sets $\{g \in \mathcal{F} : f \succ g\}$ and $\{g \in \mathcal{F} : g \succ f\}$ are open in $\mathcal{F}$.

(S3) Constant Non-Triviality. There exist $\hat{p}, \hat{q} \in \Delta(X)$ such that $\hat{p} \succ \hat{q}$, where lotteries are identified with the corresponding constant acts.

Proposition 1. *A strict partial order $\succ \subseteq \mathcal{F} \times \mathcal{F}$ satisfies (S1)–(S3) if and only if there exists a nonempty compact and convex set $\mathcal{V} \subset C(X \times S)$ and there exist $\hat{p}, \hat{q} \in \Delta(X)$ such that $U_\nu(\hat{p}) > U_\nu(\hat{q})$ for every $\nu \in \mathcal{V}$, and, for every $f, g \in \mathcal{F}$,*

$$f \succ g \iff U_\nu(f) > U_\nu(g) \quad \text{for every } \nu \in \mathcal{V}.$$

The proof, given in Appendix A, follows the same path as in Theorem 8 of Hara and Riella (2023), but removes the axiom of One-Side Monotonicity over the strict partial order. That allows the definition of an appropriate preorder to apply Theorem 1 from the previously mentioned paper.

Building on Proposition 1, we can follow Heller's (2012) approach and apply a minimax theorem to obtain a justifiable choice representation. The argument begins by constructing an auxiliary relation from the maximal menus associated with each act in the Anscombe–Aumann framework.

Theorem 1. *The following statements are equivalent:*

(1) The choice correspondence c satisfies Axioms 1–6.

(2) There exist distinct lotteries $\hat{p}, \hat{q} \in \Delta(X)$ and a nonempty compact set $\mathcal{W} \subseteq C(X \times S)$ such that, for every $A \in \mathcal{A}$,

$$c(A) = \bigcup_{\nu \in \mathcal{W}} \arg\max_{g \in A} U_\nu(g),$$

and, for every $\nu \in \mathcal{W}$, $U_\nu(\hat{p}) = 1 > 0 = U_\nu(\hat{q})$.

3.2 Multi-Prior Expected Multi-Utility Representation

We now axiomatize the multi-prior expected multi-utility representation using a choice correspondence as the primitive. Galaabaatar and Karni (2013) axiomatize this model using a strict partial order, with related approaches developed by Riella (2015) and Hara and Riella (2023). Choice-based approaches, notably Heller (2012) and Safonov (2023), do not simultaneously allow for multiple priors and multiple utility functions.

Relative to Theorem 1, we retain the same assumptions and add Strict One-Side Monotonicity to recover beliefs and state-independent tastes from the state-dependent representation. Although the auxiliary set of state-dependent utilities is convex, utilities generated by probability–utility pairs generally do not form a convex set: combining utilities with different priors and utility functions need not preserve the factorization $\nu(x, s) = \pi(s)u(x)$. As emphasized by Nau (2006) and developed by Riella (2015) and Hara and Riella (2023), the state-independent structure must instead be recovered from suitable extreme or exposed generators of the convex set of state-dependent utilities.

We adapt this logic to choice correspondences. Strict One-Side Monotonicity plays the same structural role here and, together with the axioms of Theorem 1, yields a justifiable-choice representation in terms of probability–utility pairs.

Theorem 2. *The following statements are equivalent:*

(1) The choice correspondence c satisfies Axioms 1–7.

(2) There exists a nonempty closed subset $\mathcal{M} \subseteq \Delta(S) \times C(X)$ of probability–utility pairs such that, for every menu $A \in \mathcal{A}$,

$$c(A) = \bigcup_{(\pi,u)\in\mathcal{M}} \arg\max_{g\in A} V_{\pi,u}(g).$$

Moreover, there exist $\hat{p}, \hat{q} \in \Delta(X)$ such that $\mathbb{E}_{\hat{p}}(u) > \mathbb{E}_{\hat{q}}(u)$ for every $(\pi, u) \in \mathcal{M}$.

Equivalently, Theorem 2 says that an act is chosen if and only if it can be justified by at least one probability–utility pair in the representing set. The model can therefore be interpreted as a pseudo-rational choice model in the sense of Aizerman and Malishevski (1981): the decision maker behaves as if endowed with a collection of complete Anscombe–Aumann preferences, each of which induces a probability–utility pair by satisfying Independence, Continuity, and Monotonicity, and an alternative is chosen whenever it is optimal according to at least one such preference.

Although the proof in the Appendix does not proceed entirely through this route, maximal menus suggest an alternative interpretation. For each act that is chosen from a menu, one may associate a maximal menu supporting that choice and attempt to construct a complete Anscombe–Aumann preference whose lower contour through the chosen act coincides with that maximal menu. Such a construction provides a useful way of understanding each probability–utility pair as a possible rationale behind observed choice.

The representation also shows how non-binary menus can separate indecisiveness from indifference. Let $m = \lambda f + (1 - \lambda)g$. Since each $V_{\pi,u}$ is affine, $V_{\pi,u}(m)$ lies strictly between $V_{\pi,u}(f)$ and $V_{\pi,u}(g)$ unless $V_{\pi,u}(f) = V_{\pi,u}(g)$. A pair that strictly ranks f and g therefore never makes m optimal, and m is chosen from $\{f, g, m\}$ if and only if some pair is indifferent between f and g.

Binary choice cannot tell the two cases apart: $f, g \in c(\{f, g\})$ whether they are indifferent or ranked oppositely by different pairs. Adding m to the menu does. If m is rejected, every pair has a strict ranking and the two sides disagree, so the decision maker is indecisive between f and g. If m is chosen, some pair is indifferent, though others may still disagree. The test is thus conclusive for indecisiveness and only suggestive for indifference.

4 Indecisiveness and Indifference

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5 Applications

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A Proofs

A.1 Proof of Proposition 1

Necessity. The necessity of the axioms is standard. Suppose that $\succ$ admits the stated representation. Open Continuity follows from the compactness of $\mathcal{V}$ and the continuity of U_ν. Independence follows from the affinity of each U_ν, and Constant Non-Triviality follows from the existence of $\hat{p}, \hat{q}$ such that $U_\nu(\hat{p}) > U_\nu(\hat{q})$ for every $\nu \in \mathcal{V}$. □

Sufficiency. Suppose that $\succ$ satisfies (S1)–(S3). By Constant Non-Triviality, there exist $\hat{p}, \hat{q} \in \Delta(X)$ such that $\hat{p} \succ \hat{q}$. Define the binary relation $\succsim^*$ on $\mathcal{F}$ by

$$f \succsim^* g \iff \alpha\hat{p} + (1-\alpha)f \succ \alpha\hat{q} + (1-\alpha)g \quad \text{for every } \alpha \in (0,1).$$

By the argument in the proof of Theorem 8 of Hara and Riella (2023), $\succsim^*$ is a preorder satisfying Independence, Continuity, and Directional Open-Continuity with respect to the pair $(\hat{p}, \hat{q})$. Moreover, $\succ \subseteq \succsim^*$. We may therefore apply Theorem 1 of Hara and Riella (2023). In particular, its proof yields a compact set $\mathcal{V}_0 \subset C(X \times S)$ such that

$$f \succsim^* g \iff U_\nu(f) \geq U_\nu(g) \quad \text{for every } \nu \in \mathcal{V}_0,$$

and $U_\nu(\hat{p}) - U_\nu(\hat{q}) = 1$ for every $\nu \in \mathcal{V}_0$.

We first show that $f \succ g$ implies $U_\nu(f) > U_\nu(g)$ for every $\nu \in \mathcal{V}_0$. Suppose that $f \succ g$. By Independence, $f \succ \frac{1}{2}f + \frac{1}{2}g \succ g$. Let $h = \frac{1}{2}f + \frac{1}{2}g$. By Open Continuity, there exist open neighborhoods O_f of f and O_g of g such that $f' \succ h \succ g'$ for all $(f', g') \in O_f \times O_g$. By transitivity, $f' \succ g'$ for all $(f', g') \in O_f \times O_g$.

Since the mappings $\alpha \mapsto \alpha\hat{q} + (1-\alpha)f$ and $\alpha \mapsto \alpha\hat{p} + (1-\alpha)g$ are continuous and converge to f and g, respectively, as $\alpha \to 0$, there exists $\alpha \in (0,1)$ sufficiently small such that $\alpha\hat{q} + (1-\alpha)f \in O_f$ and

$\alpha\hat{p}+(1-\alpha)g \in O_g$. Consequently, $\alpha\hat{q}+(1-\alpha)f \succ \alpha\hat{p}+(1-\alpha)g$. Since $\succ \subseteq \succsim^*$, the representation of $\succsim^*$ implies that, for every $\nu \in \mathcal{V}_0$, $\alpha U_\nu(\hat{q})+(1-\alpha)U_\nu(f) \geq \alpha U_\nu(\hat{p})+(1-\alpha)U_\nu(g)$. Rearranging, $U_\nu(f)-U_\nu(g) \geq \frac{\alpha}{1-\alpha}\left[U_\nu(\hat{p})-U_\nu(\hat{q})\right]$. By the normalization obtained in the proof of Theorem 1 of Hara and Riella (2023), we have $U_\nu(\hat{p})-U_\nu(\hat{q})=1$ for every $\nu \in \mathcal{V}_0$. Therefore,

$$U_\nu(f)-U_\nu(g) \geq \frac{\alpha}{1-\alpha} > 0 \quad \text{for every } \nu \in \mathcal{V}_0.$$

Hence, $f \succ g$ implies $U_\nu(f) > U_\nu(g)$ for every $\nu \in \mathcal{V}_0$.

We now prove the converse implication. Suppose that $f, g \in \mathcal{F}$ satisfy $U_\nu(f) > U_\nu(g)$ for every $\nu \in \mathcal{V}_0$. Since $\mathcal{V}_0$ is compact and the map $\nu \mapsto U_\nu(f)-U_\nu(g)$ is continuous,

$$m := \min_{\nu \in \mathcal{V}_0}\big[U_\nu(f)-U_\nu(g)\big] > 0.$$

Similarly, define $M := \max_{\nu \in \mathcal{V}_0}\big[U_\nu(\hat{p})-U_\nu(\hat{q})\big]$. By the normalization obtained from Theorem 1, $M=1$. Choose $\alpha \in (0,1)$ sufficiently small such that $(1-\alpha)m > \alpha M$. Then, for every $\nu \in \mathcal{V}_0$, the affinity of U_ν implies

$$\begin{aligned} &U_\nu\big(\alpha\hat{q}+(1-\alpha)f\big)-U_\nu\big(\alpha\hat{p}+(1-\alpha)g\big) \\ &\quad = (1-\alpha)\big[U_\nu(f)-U_\nu(g)\big]-\alpha\big[U_\nu(\hat{p})-U_\nu(\hat{q})\big] \\ &\quad \geq (1-\alpha)m-\alpha M > 0. \end{aligned}$$

Hence, by the representation of $\succsim^*$, $\alpha\hat{q}+(1-\alpha)f \succsim^* \alpha\hat{p}+(1-\alpha)g$. By the definition of $\succsim^*$, for every $\beta \in (0,1)$,

$$\begin{aligned} &\beta\hat{p}+(1-\beta)\big[\alpha\hat{q}+(1-\alpha)f\big] \\ &\qquad \succ \beta\hat{q}+(1-\beta)\big[\alpha\hat{p}+(1-\alpha)g\big]. \end{aligned}$$

In particular, let $\beta = \frac{\alpha}{1+\alpha}$. Since $\beta = (1-\beta)\alpha$, simplifying the preceding comparison gives

$$\begin{aligned} &\frac{2\alpha}{1+\alpha}\left(\frac{\hat{p}+\hat{q}}{2}\right)+\frac{1-\alpha}{1+\alpha}f \\ &\qquad \succ \frac{2\alpha}{1+\alpha}\left(\frac{\hat{p}+\hat{q}}{2}\right)+\frac{1-\alpha}{1+\alpha}g. \end{aligned}$$

Since $\frac{1-\alpha}{1+\alpha} \in (0,1)$, Independence implies $f \succ g$. Therefore, $f \succ g \iff U_\nu(f) > U_\nu(g)$ for every $\nu \in \mathcal{V}_0$.

It remains to show that $\mathcal{V}$ can be chosen convex. Let $\mathcal{V}_0$ be the compact representing set obtained above and define $\mathcal{V} := \overline{\text{co}}(\mathcal{V}_0)$, where the closure is taken in the supremum norm. Since $C(X \times S)$ is a Banach space, the closed convex hull of a compact set is compact. Moreover, for any $f \succ g$, continuity and compactness imply $m := \min_{\nu \in \mathcal{V}_0}[U_\nu(f)-U_\nu(g)] > 0$. By linearity in ν, this lower bound is preserved under convex combinations and, by continuity, under closure. Thus $U_\nu(f) > U_\nu(g)$ for every $\nu \in \mathcal{V}$. The converse follows from $\mathcal{V}_0 \subseteq \mathcal{V}$. Hence, convexification preserves the representation.

□

A.2 Proof of Theorem 1

Sufficiency. Suppose that Axioms 1–6 hold. For any $f, g \in \mathcal{F}$, define the relation $\succ_c$ by $f \succ_c g \iff g \notin c(\{f, g\})$.

Claim 1. *The relation $\succ_c$ is a strict partial order that satisfies axioms (S1)–(S3).*

Proof. Nonemptiness of $c(A)$ for every $A \in \mathcal{A}$, together with Pseudo-WARP, implies that $\succ_c$ is a strict partial order. Independence of $c(\cdot)$ implies that $\succ_c$ satisfies Independence. Constant Non-Triviality of $c(\cdot)$ implies Constant Non-Triviality of $\succ_c$: there exist $\hat{p}, \hat{q} \in \Delta(X)$ such that $\hat{q} \notin c(\{\hat{p}, \hat{q}\})$, so $\hat{p} \succ_c \hat{q}$. Similarly, Continuity of $c(\cdot)$ implies Open Continuity of $\succ_c$. Therefore, $\succ_c$ satisfies all the properties required to apply Proposition 1. □

Now fix $f \in \mathcal{F}$ and $A \in \mathcal{A}$ such that $f \in c(A)$.

Claim 2. *For every $A \in \mathcal{A}$ and $f \in c(A)$, the set $\{B \in \mathcal{A} : A \subseteq B \text{ and } f \in c(B)\}$ has a maximal element, M, with respect to set inclusion.*

Proof. Suppose $\mathcal{B}$ is a subset of $\{B \in \mathcal{A} : A \subseteq B \text{ and } f \in c(B)\}$ linearly ordered by $\supseteq$. Let $\bar{B} := \mathrm{cl}(\bigcup \mathcal{B})$. This implies that there exists a sequence (B^m) of finite subsets of $\bar{B}$ such that $B^m \to \bar{B}$. By the definition of $\bar{B}$, we can find such a sequence with $B^m \subseteq \bigcup \mathcal{B}$ for every $m \in \mathbb{N}$. Finally, since each B^m is finite and $\mathcal{B}$ is linearly ordered by $\supseteq$, we can find, for each $m \in \mathbb{N}$, a set $\bar{B}^m \in \mathcal{B}$ with $B^m \subseteq \bar{B}^m$. But then, $\bar{B}^m \to \bar{B}$ and $f \in c(\bar{B}^m)$ for every $m \in \mathbb{N}$. Continuity implies that $f \in c(\bar{B})$. That is, $\bar{B}$ is an upper bound for $\mathcal{B}$. Now Zorn's lemma implies that the collection $\{B \in \mathcal{A} : A \subseteq B \text{ and } f \in c(B)\}$ has a maximal element. □

Claim 3. *Both M and $\mathcal{F} \setminus M$ are convex.*

Proof. Since $f \in c(M)$, if $g, h \in M$ and $\alpha \in (0, 1)$, No-Hedging and maximality of M immediately imply that $\alpha g + (1 - \alpha)h \in M$. Thus, M is convex.

To prove that $\mathcal{F} \setminus M$ is convex, suppose that $g \notin M$ and $h \notin M$, and let $\alpha \in (0, 1)$. By maximality of M, $f \notin c(M \cup \{g\})$ and $f \notin c(M \cup \{h\})$. Therefore, $f \notin c(M \cup \{g\}) \cup c(M \cup \{h\})$. Indirect Ambiguity Aversion then implies that $f \notin c\big(M \cup \{\alpha g + (1-\alpha)h\}\big)$. Since $f \in c(M)$, it follows that $\alpha g + (1 - \alpha)h \notin M$. Hence, $\mathcal{F} \setminus M$ is convex. □

By Claim 1 and Proposition 1, let $\mathcal{V} \subseteq C(X \times S)$ be a nonempty compact and convex set representing $\succ_c$. We may normalize this family so that $U_\nu(p_*) = 0$, $U_\nu(p^*) = 1$ $(\nu \in \mathcal{V})$. Indeed, $p^* \succ_c p_*$, so the normalizing denominators are positive and bounded away from zero on the compact family. Normalize each utility and take the closed convex hull again; compactness and the strict binary representation are preserved. Applying Best and Worst to singleton menus and letting the mixture weight on the endpoint tend to zero gives $0 \le U_\nu(g) \le 1$ $(g \in \mathcal{F}, \nu \in \mathcal{V})$.

By Claim 2, for any fixed $f \in c(A)$, there exists M such that $f \in c(M)$ and $A \subseteq M$. Pseudo-WARP implies that, for every $g \in M$, $f \in c(\{f, g\})$. Therefore,

$$\begin{aligned} f \in c(A) &\implies \forall g \in M, \quad g \not\succ_c f \\ &\implies \forall g \in M,\ \exists \nu \in \mathcal{V} : \quad U_\nu(f) \ge U_\nu(g) \\ &\iff \min_{g \in M} \max_{\nu \in \mathcal{V}} \big[U_\nu(f) - U_\nu(g)\big] \ge 0. \end{aligned}$$

Since M and $\mathcal{V}$ are compact and convex, and $U_\nu(f)-U_\nu(g)$ is continuous and affine in each variable, Sion's Minimax Theorem gives

$$\min_{g\in M}\max_{\nu\in\mathcal{V}}\big[U_\nu(f)-U_\nu(g)\big]=\max_{\nu\in\mathcal{V}}\min_{g\in M}\big[U_\nu(f)-U_\nu(g)\big].$$

It follows that $f\in c(A)\implies\exists\nu\in\mathcal{V},\ \forall g\in M,\ U_\nu(f)\geq U_\nu(g)$. Consequently,

$$c(A)\subseteq\bigcup_{\nu\in\mathcal{V}}\arg\max_{g\in A}U_\nu(g).$$

The reverse inclusion need not hold for the whole convex family $\mathcal{V}$. We next select utilities that justify the original choice without introducing additional choices on any other menu.

Fix a constant act q whose lottery has full support on X. Such a lottery exists because X is compact metrizable: assign positive probabilities to the points of a countable dense subset. For a fixed $\lambda\in(0,1)$, put $T_\lambda(g)=\lambda g+(1-\lambda)q$, $f_\lambda=T_\lambda(f)$, $B_\lambda=T_\lambda(M)$. Independence gives $f_\lambda\in c(B_\lambda)$. Let M_λ be a maximal compact menu containing B_λ and choosing f_λ. By the preceding claims, both M_λ and its complement are convex. Moreover,

$$M_\lambda\cap T_\lambda(\mathcal{F})=T_\lambda(M).\tag{1}$$

To prove this, write the intersection as $T_\lambda(B)$. Since T_λ is a homeomorphism onto its compact image, B is compact and contains M. Pseudo-WARP implies contraction of choice: if $C\subseteq D$ and $g\in c(D)\cap C$, then $g\in c(C)$. Thus $f_\lambda\in c(T_\lambda(B))$, and Independence gives $f\in c(B)$. Maximality of M implies $B=M$.

Apply the preceding minimax argument to M_λ and f_λ. There is $\nu\in\mathcal{V}$ such that, writing $a=U_\nu(f_\lambda)$,

$$M_\lambda\subseteq\{g\in\mathcal{F}:U_\nu(g)\leq a\}.\tag{2}$$

Every $f_\lambda(s)$ has full support. Since U_ν is nonconstant,

$$\min_{g\in\mathcal{F}}U_\nu(g)<a<\max_{g\in\mathcal{F}}U_\nu(g).$$

Indeed, the extrema are the sums of the statewise extrema of $\nu(\cdot,s)$; integration against a full-support probability is strictly between the extrema of each nonconstant continuous coordinate. In particular, $0<a<1$.

We claim that equality holds in (2). Suppose first that $g\notin M_\lambda$ and $U_\nu(g)<a$. Probabilities dominated by a finite multiple of a full-support probability μ are weakly dense in $\Delta(X)$. To see this, approximate a probability by a finite-support probability and replace each point mass at x by $\mu(\cdot\cap B(x,\epsilon))/\mu(B(x,\epsilon))$. The resulting probabilities have bounded densities with respect to μ and approximate the point masses as $\epsilon\downarrow 0$. Applying this in every state, and using the openness of $\mathcal{F}\setminus M_\lambda$, we can choose $\tilde{g}\notin M_\lambda$ such that $U_\nu(\tilde{g})<a$ and $\tilde{g}(s)\leq Cf_\lambda(s)\qquad(s\in S)$ for some finite C. For sufficiently small $t>0$, $h=(1+t)f_\lambda-t\tilde{g}$ is therefore an act, and $U_\nu(h)=a+t[a-U_\nu(\tilde{g})]>a$. By (2), $h\notin M_\lambda$. But $f_\lambda=\frac{t}{1+t}\tilde{g}+\frac{1}{1+t}h$, contradicting convexity of the complement. Hence every exterior act has utility at least a. If $g\notin M_\lambda$ had $U_\nu(g)=a$, then $(1-t)g+tp_*$ would remain outside the closed menu M_λ for sufficiently small $t>0$, while its utility would be $(1-t)a<a$.

This is impossible. We have proved

$$M_\lambda = \{g \in \mathcal{F} : U_\nu(g) \leq a\}. \tag{3}$$

Combining this identity with (1) and affinity yields

$$M = \{g \in \mathcal{F} : U_\nu(g) \leq U_\nu(f)\}.$$

Next, every act g with $U_\nu(g) = a$ is chosen from M_λ. For $t \in (0,1)$, put $g_t = (1-t)g + tp^*$. Since $U_\nu(g_t) > a$, maximality implies $f_\lambda \notin c(M_\lambda \cup \{g_t\})$. If g_t were also unchosen, then $c(M_\lambda \cup \{g_t\}) \subseteq M_\lambda$; Pseudo-WARP and $f_\lambda \in c(M_\lambda)$ would give a contradiction. Thus $g_t \in c(M_\lambda \cup \{g_t\})$, and Continuity gives

$$\{g : U_\nu(g) = a\} \subseteq c(M_\lambda). \tag{4}$$

Now let $B \in \mathcal{A}$ and $g \in \arg\max_{h \in B} U_\nu(h)$. Choose $\alpha \in (0,1]$ and $j \in \{p^*, p_*\}$ so that $U_\nu(\alpha g + (1-\alpha)j) = a$. If $U_\nu(g) > a$, take $j = p_*$ and $\alpha = a/U_\nu(g)$; if $U_\nu(g) < a$, take $j = p^*$ and $\alpha = (1-a)/(1-U_\nu(g))$; otherwise take $\alpha = 1$. By (3), $\alpha B + (1-\alpha)j \subseteq M_\lambda$. Equation (4), contraction, and Independence imply $g \in c(B)$. Thus the selected ν satisfies $\arg\max_{g \in B} U_\nu(g) \subseteq c(B)$ for every $B \in \mathcal{A}$. It also makes the original f optimal on $A \subseteq M$.

Define

$$\mathcal{W} = \left\{ \nu \in \mathcal{V} : \arg\max_{g \in B} U_\nu(g) \subseteq c(B) \text{ for every } B \in \mathcal{A} \right\}.$$

For every A and $f \in c(A)$, the construction supplies such a $\nu \in \mathcal{W}$ justifying f. Consequently $\mathcal{W}$ is nonempty and

$$c(A) = \bigcup_{\nu \in \mathcal{W}} \arg\max_{g \in A} U_\nu(g).$$

To verify compactness, suppose $\nu_n \in \mathcal{W}$ and $\nu_n \to \nu \in \mathcal{V}$ in the supremum norm. Fix $B \in \mathcal{A}$ and $g \in \arg\max_{h \in B} U_\nu(h)$. If $U_\nu(g) < 1$, then for fixed $t > 0$ the act $g_t = (1-t)g + tp^*$ has strictly higher utility than every member of B. Uniform convergence implies the same inequality under U_{ν_n} for all sufficiently large n. Since $\nu_n \in \mathcal{W}$, $g_t \in c(B \cup \{g_t\})$. Letting $t \downarrow 0$ and using Continuity gives $g \in c(B)$. If $U_\nu(g) = 1$, first replace B, g by $\frac{1}{2}B + \frac{1}{2}p_*, \frac{1}{2}g + \frac{1}{2}p_*$, apply the preceding case, and use Independence. Hence $\nu \in \mathcal{W}$, so $\mathcal{W}$ is closed in the compact set $\mathcal{V}$ and therefore compact. The common normalization gives the required endpoints $\hat{p} = p^*$ and $\hat{q} = p_*$. This completes the sufficiency argument. □

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